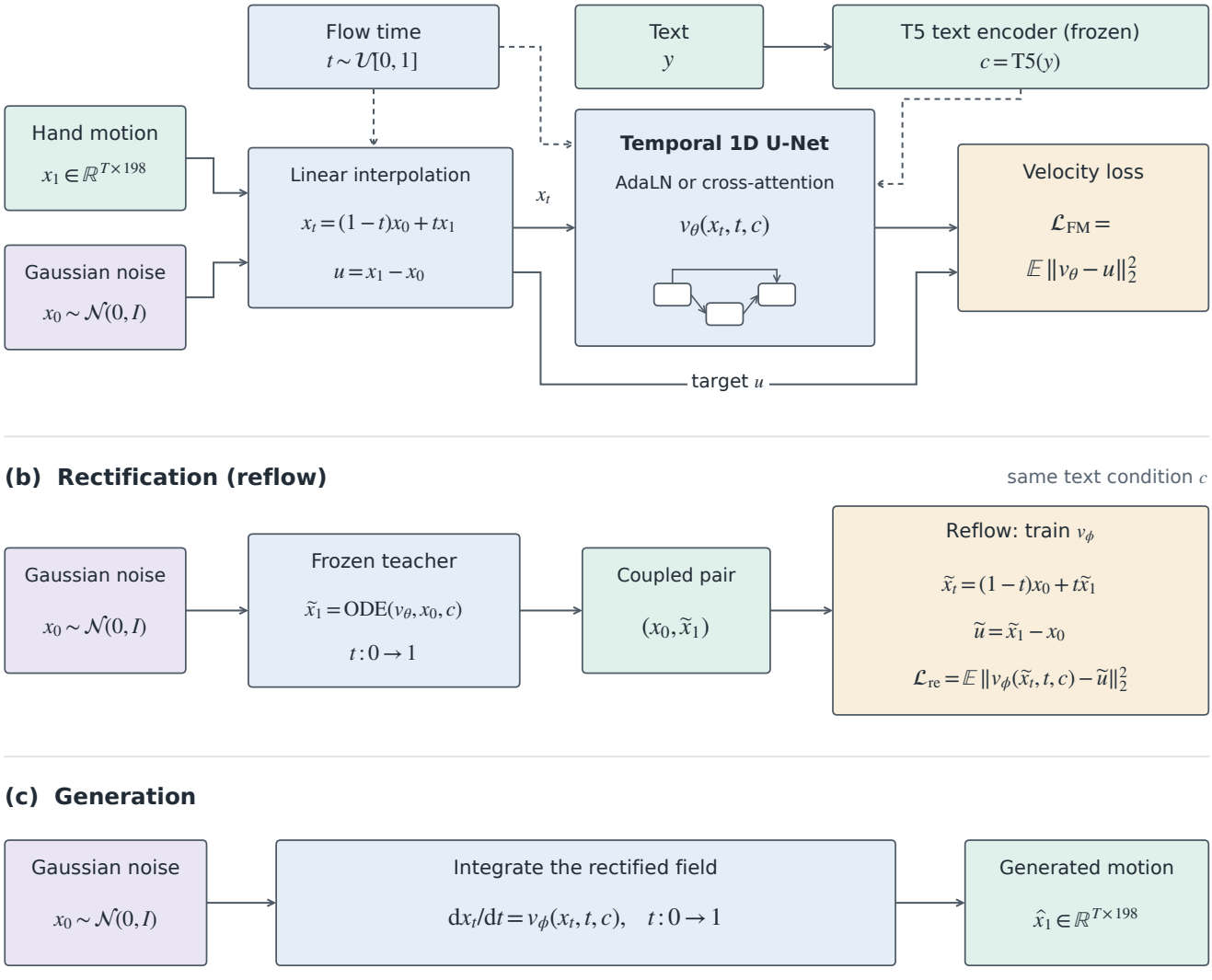
(a) Conditional flow matching in motion space
Flow time
$t \sim \mathcal{U}[0,1]$
Text
y
T5 text encoder (frozen)
$c = \mathrm{T5}(y)$
Hand motion
$x_1 \in \mathbb{R}^{T\times 198}$
Linear interpolation
$x_t = (1-t)x_0 + tx_1$
$u = x_1 - x_0$
Gaussian noise
$x_0 \sim \mathcal{N}(0, I)$
x_t
Temporal 1D U-Net
AdaLN or cross-attention
$v_\theta(x_t, t, c)$
Velocity loss
$\mathcal{L}_{\mathrm{FM}} =$
$\mathbb{E}\,\|v_\theta - u\|_2^2$
target u
(b) Rectification (reflow)
same text condition c
Gaussian noise
$x_0 \sim \mathcal{N}(0, I)$
Frozen teacher
$\tilde{x}_1 = \mathrm{ODE}(v_\theta, x_0, c)$
$t: 0 \to 1$
Coupled pair
$(x_0, \tilde{x}_1)$
Reflow: train v_ϕ
$\tilde{x}_t = (1-t)x_0 + t\tilde{x}_1$
$\tilde{u} = \tilde{x}_1 - x_0$
$\mathcal{L}_{\mathrm{re}} = \mathbb{E}\,\|v_\phi(\tilde{x}_t, t, c) - \tilde{u}\|_2^2$
(c) Generation
Gaussian noise
$x_0 \sim \mathcal{N}(0, I)$
Integrate the rectified field
$\mathrm{d}x_t/\mathrm{d}t = v_\phi(x_t, t, c), \quad t: 0 \to 1$
Generated motion
$\hat{x}_1 \in \mathbb{R}^{T\times 198}$